

Dynamic-Payload Pick-and-Pour Control for a UR5 Barista Robot: Progress Report

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I. INTRODUCTION

A. Problem Description

B. Related Work

UR5 modelling. Kebria et al. [1] provide a complete set of kinematic and dynamic models for the UR5, including DH parameters, closed-form forward and inverse kinematics, the inertia, Coriolis, and gravity terms from the Lagrangian formulation, and a SimMechanics model. They validate the model with a PD controller using feedback linearization. Their dynamics assume a fixed payload, which is the assumption we relax, and their model forms the basis of our kinematic and dynamic equations in Section II.

Vision-based liquid volume estimation. Schober et al. [2] estimate the liquid volume in transparent laboratory containers from a single RGB image using a two-step CNN, then execute pouring trajectories selected from a physics simulation on a UR5e. Their pouring strategy rotates the container about a fixed liquid exit point, which suits small openings. Their end-to-end errors were large (RMSE of 21–26 mL for 30–50 mL targets), and the authors note that volume-estimation and simulation errors accumulate, making the system unsuitable for precise work. This motivates adding a direct measurement of the poured mass, which we obtain from a scale, instead of relying on vision alone.

Force/torque perception of liquids. Luo et al. [3] use a wrist-mounted force/torque sensor on a UR5e to push a whisk through pancake batter, inferring uniformity, liquid level, and water–flour ratio from the resistance torque. They then pour with open-loop control at a constant wrist angular speed and an MLP that sets the arm speed to achieve a desired stroke width. Their work shows that a wrist force /torque sensor carries useful information about the liquid, but their pouring is open loop and uses a single fixed pouring motion, so it does not adapt to the measured amount poured.

Learned pouring policies. Babaiaans et al. [4] train PourNet, a reinforcement-learning policy (PPO with an intrinsic curiosity module and curriculum learning) in a Unity3D/NVIDIA Flex simulator, and execute the resulting end-effector motion through a nonlinear MPC planner on a Kinova Move. A force/torque sensor on the second arm measures the weight in the target container, and the policy generalizes across liquids and containers without real-world fine-tuning, with deviations

TABLE I
COMPARISON OF RELATED WORK AND THIS PROJECT

Work	Sensing	Pour control	Time-varying load modelled
[1]	–	PD tracking	No
[2]	RGB camera	Simulation-selected	No
[3]	Wrist F/T	Open-loop	No
[4]	F/T (weight)	RL + NMPC	Learned implicitly
Ours	Camera, scale, wrist F/T	Model-based, dynamic load	Yes

of roughly 2–9 g. The approach needs large amounts of simulated training and treats the pour as an end-effector planning problem rather than analyzing the arm’s own kinematics and dynamics under a changing load.

Comparison and our approach. Table I summarizes how these works differ from ours. Vision-based approaches [2] are flexible but imprecise, haptic approaches [3] perceive liquid properties but pour open loop, and learned policies [4] achieve good accuracy but need extensive training and do not model the arm’s changing dynamics. Detailed UR5 models [1] exist but assume a constant payload. Our work uses the UR5 model from [1] and extends it with a time-varying payload mass. We fuse a camera, a scale, and a wrist force/torque sensor for feedback, and we use a model-based controller that compensates for changing load. We simulate the full pick-up-and-pour sequence in ROS2 and Gazebo and study manipulability and singularities along the trajectory.

II. METHODOLOGY

A. Robot Sequence

B. Kinematic Model

C. Dynamic Model with Time-Varying Payload

D. Controller

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